Lesson notes · for you

Standard deviation and outliers

Data science · Level 2 · Lesson 2

Time	Materials
30 min	A notebook and a pencil; A phone or tablet with Darsi, for the lab

Learning goals

- Read a standard deviation.
- Flag outliers with the IQR rule.
- Check outliers before acting.

Activity: Outlier detectives

1. Warm-up (5 min): Compare two lists with the same mean but very different spreads.
2. Find (10 min): With a partner, work out the quartiles and the fences for a list of journey times.
3. Investigate (10 min): For each outlier, decide what could explain it and how to check.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

Discussion

- What does a big standard deviation mean?
- Why not delete outliers?
- Which outlier surprised you?

Check yourself

- Q1 is 20 and Q3 is 30. Write the two fences.

Try it this week

Write down how long your journeys take for a week, then find the values that stand out. Is each one a mistake, or something real?

Worksheet

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Name _____ Class _____ Date _____

1 **Class C has a standard deviation of 2, class D of 6. Which has more mixed marks?**

◯ Class D ◯ They are the same ◯ Class C

2 **Two lists with the same mean always have the same spread.**

◯ True ◯ False

3 **Print every mark more than two standard deviations above the mean.**

```
from ulab import numpy as np
m = np.array([11, 12, 12, 13, 12, 30])
avg = np.mean(m)
sd = np.std(m)
print(avg)
```

Write your program:

4 **Use the IQR rule.**

Q1 = 10 and Q3 = 14, so the IQR is _____ and the upper fence is _____.

Word bank: 18 · 20 · 4 · 6

5 **A journey time of 300 minutes appears in your data. What do you do first?**

◯ Delete it ◯ Check whether it is a mistake or real ◯ Use it as the average

✎ **Think of a time when you were the outlier. What would an average have hidden about you?**

Check your answers

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1 **Class D**

A larger standard deviation means marks lie further from the mean.

2 **False**

11, 12, 13 and 5, 12, 19 share a mean of 12 but differ hugely in spread.

3

```
from ulab import numpy as np
m = np.array([11, 12, 12, 13, 12, 30])
avg = np.mean(m)
sd = np.std(m)
for x in m:
    if x > avg + 2 * sd:
        print(int(x))
```

The mean is 15 and the SD about 6.7, so the cut-off is about 28.5.

4 **Q1 = 10 and Q3 = 14, so the IQR is 4 and the upper fence is 20.**

IQR = 14 − 10 = 4, and 14 + 1.5 × 4 = 20.

5 **Check whether it is a mistake or real**

An outlier is a question: check it before deciding.